

ARM7 Architecture

1 ARM7 Overview

ARM7 belongs to the **Advanced RISC Machine (ARM)** family and follows **Reduced Instruction Set Computer (RISC)** principles, emphasizing simplicity, speed, and low power consumption.

Key Features

- Power-efficient 32-bit microprocessor
- Supports both 32-bit ARM and 16-bit Thumb instructions
- Register-oriented load-store architecture
- Multiple operating modes
- Simple design with low gate count

Applications

- Wireless and mobile communication
- Embedded and real-time systems
- Portable instruments and smart cards

Thumb Instruction Set

Thumb mode uses 16-bit instructions to improve code density, reduce memory usage, and lower power consumption while maintaining performance.

2 Debugging and Instruction Execution

Debug Support

- **JTAG Debug Port** (Joint Test Action Group)
- **Embedded ICE** (In-Circuit Emulator)

These features allow in-system debugging, program download directly into the chip, breakpoints, watchpoints, and step-by-step execution.

Instruction Execution Characteristics

- Single data bus for both code and data
- Register-to-register instructions execute in one clock cycle
- Load and store instructions require multiple cycles

Memory access is slower than register access, which is why load/store instructions take multiple cycles.

3 Pipeline Architecture

ARM7 improves performance using a **3-stage instruction pipeline**.

Pipeline Stages

1. **Fetch** – Instruction is fetched from memory
2. **Decode** – Instruction is interpreted and operands identified
3. **Execute** – Operation is performed

Each pipeline stage has independent hardware, allowing up to three instructions to be processed simultaneously.

Pipeline Advantage

- Improves instruction throughput
- Keeps CPU hardware continuously busy

4 Programming Model and Register Set

ARM7 provides **16 CPU registers**.

Register Classification

Register	Function
R0–R12	General-purpose registers
R13	Stack Pointer (SP)
R14	Link Register (LR)
R15	Program Counter (PC)

Stack and Procedure Handling

- No dedicated PUSH or POP instructions
- Stack handled using load/store multiple instructions
- Return address stored in Link Register during procedure calls

Nested subroutine calls require saving the Link Register onto the stack to prevent over-writing return addresses.

5 Current Program Status Register (CPSR)

CPSR stores the **current state and control information** of the processor.

Functions of CPSR

- Holds condition code flags
- Controls processor mode
- Enables or disables interrupts
- Selects ARM or Thumb execution state

Condition Code Flags

Flag	Meaning
N	Negative result
Z	Zero result
C	Carry generated
V	Overflow generated

6 Control and Mode Bits

Interrupt Control

- **I bit** – Disable IRQ interrupts
- **F bit** – Disable FIQ interrupts

Instruction State

- **T = 0** – ARM instruction (32-bit)
- **T = 1** – Thumb instruction (16-bit)

Processor Operating Modes

Mode	Binary Code
User	10000
FIQ	10001
IRQ	10010
Supervisor	10011
Abort	10111
Undefined	11011
System	11111

7 Quick Summary

ARM7 is a low-power, high-performance RISC processor featuring pipelining, efficient register usage, strong debugging support, and flexible execution control through CPSR.